

MIDTERM EXAM

IT SERVICE MANAGEMENT

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You will make it. Good luck!

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CHALLENGE 1

Closed book, no GenAI is allowed.

Complete all the questions and take a screenshot after submitting your answers.

The link to challenge 1: <https://forms.gle/i79vTJLUDSDrkz4n6>

The screenshot shows a Google Forms interface for a 'Mid Exam - IT Service Management'. The URL in the browser's address bar is https://docs.google.com/forms/d/e/1FAIpQLSct9rONBzh_aCvPB88POLSOFiH2x-wiEZq75w_TKYpznJPHg/viewscore?viewscore=AE0zAgCcq_p5Gw.... The form title is 'Mid Exam - IT Service Management'. Below the title, it says 'Total points 47/50'. The section is labeled 'PART I. MULTIPLE-CHOICES'. A message states: 'You are given 40 minutes to answer 50 multiple-choice questions. Other details are as follows.' A bulleted list contains the following rules:

- This is a closed-book test.
- You are NOT allowed to use any other assistance, including GenAI.
- One attempt only.
- All questions are shuffled by default.
- You are required to answer all the questions.

Below the list, it says: 'Whenever you are done, you can close the window and start working on the second part of the exam.' This is followed by 'Good luck!' and 'Best wishes, Hanim Astuti'. A note at the bottom states: 'The respondent's email (ryanfebriana899@gmail.com) was recorded on submission of this form.' On the right side of the form, there is a purple button that says '0 of 0 points'. At the bottom of the form, there is a field labeled 'Your Class *' and a small icon for 'MITI R (Rendani)'.

CHALLENGE 2

Closed book, no GenAI is allowed.

Scenario

You are an IT Service Analyst at Bank Nusantara Digital, an Indonesian banking company. The management suspects that several helpdesk tickets exported from ServiceNow contain incomplete, inconsistent, or illogical information.

Your task is to audit the dataset, identify data-quality issues, and recommend improvements to strengthen reporting accuracy.

Dataset Information (Find the dataset on the next page)

- Each row represents one helpdesk record. Columns:
Ticket ID, Ticket Type, Category, Priority, Resolution Time (hrs),
Description (Summary), Department, Ticket Agent, Ticket User
- Example of errors: Missing data, unrealistic priorities, etc.

Your Tasks

- Review the errors and improve.
- Share the link to your (improved excel)
- And copy and paste your table here in this document.
- Answer this:
 1. Analyze the dataset and identify as many errors as possible.
 2. What did you do to improve the errors?
 3. What do you think a good service desk dataset consist of?
 4. Create a one or two paragraphs to report to your manager.

The Dataset

Ticket ID	Ticket Type	Category	Priority	Resolution Time (hrs)	Description (Summary)	Department	Ticket Agent	Ticket User
INC001	Request	Network	P1	3	VPN connection lost for Finance user	Corporate Finance		Agus
REQ002	Incident	Hardware	P3	5	New email account for new intern	HR		Tono
INC003	Incident	Software	P2		Core banking app crashes during batch run	Digital Services	Budi	Tono
REQ004	Request			3	Laptop replacement for senior officer	Operations	Dewi	Lina
INC005		Hardware	P1	0	Printer not responding in teller area	Retail Banking	Dewi	
INC006	Incident	Access		145	Password reset request from Marketing		Dewi	Lina
REQ007	Request	Network	P1	8	Install Microsoft Office for new staff	HR	Budi	Tono
INC008		Network	P2	12	Internet down at Denpasar branch	Operations	Budi	Fajar
REQ009	Request	Software	P3		Setup new scanner at Bandung office	Retail Banking	Andi	Rina
INC010	Incident	Software	P5	1	System slow during transaction posting	Digital Services	Andi	Rina
REQ011	Request	Access		6	Grant access to credit-risk portal	Corporate Finance	Eko	Sari
INC012	Incident	Hardware	P1	210	Server overheating in data center		Dewi	Lina
REQ013	Incident	Hardware	P3	9	Request new mouse for employee	HR	Dewi	
INC014	Incident		P2	14	ATM network outage in Surabaya branch	Retail Banking	Dewi	Rina
REQ015		Software	P4		Install SPSS for Risk Analytics team		Andi	Tono

Link to your improved excel:

https://docs.google.com/spreadsheets/d/1KDol4bbTOK96EU7Hyl1DX49UIbz_zSmtVedtO1apXX8/edit?gid=159089693#gid=159089693

Copy and paste your table:

Ticket ID	Ticket Type	Category	Priority	Resolution Time (hrs)	Description (Summary)	Department	Ticket Agent	Ticket User
INC001	Incident	Network	P1	3	VPN connection lost for Finance user	Corporate Finance	Maria	Agus
REQ002	Request	Software	P3	5	New email account for new intern	HR	Andre	Tono
INC003	Incident	Software	P2	3	Core banking app crashes during batch run	Digital Services	Budi	Tono
REQ004	Request	Hardware	P3	10	Laptop replacement for senior officer	Operations	Dewi	Lina
INC005	Incident	Hardware	P1	2	Printer not responding in teller area	Retail Banking	Dewi	Rina
REQ006	Request	Access	P4	0.25	Password reset request from Marketing	Marketing	Dewi	Lina
REQ007	Request	Software	P3	8	Install Microsoft Office for new staff	HR	Budi	Tono
INC008	Incident	Network	P2	4	Internet down at Denpasar branch	Operations	Budi	Fajar
REQ009	Request	Hardware	P3	6	Setup new scanner at Bandung office	Retail Banking	Andi	Rina
INC010	Incident	Software	P2	4	System slow during transaction posting	Digital Services	Andi	Rina
REQ011	Request	Access	P3	6	Grant access to credit-risk portal	Corporate Finance	Eko	Sari

INC012	Incident	Hardware	P1	1.5	Server overheating in data center	IT Infrastructure	Dewi	Lina
REQ013	Request	Hardware	P4	4	Request new mouse for employee	HR	Dewi	Tono
INC014	Incident	Network	P1	3	ATM network outage in Surabaya branch	Retail Banking	Dewi	Rina
REQ015	Request	Software	P4	6	Install SPSS for Risk Analytics team	Corporate Finance	Andi	Tono

Note: Cell berwarna kuning berarti bagian yang diimprove dari dataset original

Your answers to the three questions:

1. Analyze the dataset and identify as many errors as possible.

JAWAB:

Berdasarkan analisis per baris, saya mengidentifikasi beberapa jenis kesalahan pada dataset:

- **Inkonsistensi Tipe Tiket dan ID Tiket:** Terdapat tiket dengan ID INC (Incident) tetapi memiliki Tipe Tiket Request (REQ002) dan sebaliknya (INC006). Ini mengindikasikan ketidaksesuaian antara ID yang diberikan oleh sistem dan tipe yang dicatat.
- **Waktu Resolusi yang Tidak Logis:** Beberapa tiket memiliki waktu resolusi yang sangat tidak masuk akal. Contohnya, tiket "Password reset" dengan waktu 145 jam (INC006) atau "Server overheating" dengan 210 jam (INC012). Nilai-nilai ini jauh dari standar operasional yang diharapkan.
- **Prioritas yang Tidak Sesuai:** Prioritas yang ditetapkan sering kali tidak mencerminkan urgensi masalah. Contohnya, tiket "Internet down" memiliki prioritas P2 (INC008), padahal ini seharusnya P1. Sebaliknya, tiket "Install Microsoft Office"

memiliki prioritas P1 (REQ007), padahal ini adalah permintaan rutin.

- **Data Hilang (Missing Data):** Banyak kolom yang kosong, termasuk Tipe Tiket, Kategori, Prioritas, Waktu Resolusi, dan Departemen. Contohnya, tiket REQ004, INC005, INC006, REQ009, INC010, REQ011, INC012, INC014, dan REQ015 memiliki beberapa sel yang tidak terisi.

2. What did you do to improve the errors?

JAWAB:

Tindakan yang saya lakukan untuk memperbaiki kesalahan tersebut adalah:

- **Koreksi Tipe Tiket dan Prioritas:** Saya mencocokkan kembali Tipe Tiket dan Prioritas dengan deskripsi tiket untuk memastikan konsistensi. Misalnya, "Password reset" yang sebelumnya Incident dan P3 diubah menjadi Request dan P4.
- **Memperbaiki ID Menyesuaikan Deskripsi Dataset:** Dengan menjadikan deskripsi dataset sebagai patokan untuk menentukan tipe tiket, saya juga memperbaiki ID untuk incident dimulai dengan "INC" sedangkan untuk request dimulai dengan "REQ".
- **Mengisi Data yang Hilang:** Saya mengisi sel-sel yang kosong berdasarkan konteks yang ada di kolom lain. Contohnya, Kategori untuk "Setup new scanner" diisi Hardware dan Departemen untuk "Install SPSS" diisi Corporate Finance atau Risk Management. Lalu untuk ticket agent dan ticket user juga saya berlakukan pengisian secara dummy.
- **Normalisasi Waktu Resolusi:** Untuk nilai yang tidak logis, saya mengisinya dengan angka yang lebih masuk akal berdasarkan prioritas dan tipe tiket. Waktu resolusi yang terlalu lama seperti **145 jam** atau **210 jam** diubah menjadi nilai yang realistis, seperti **0.25** atau **1.5** jam.

3. What do you think a good service desk dataset consist of?

JAWAB:

Dataset *helpdesk* yang baik harus memenuhi kriteria berikut:

- **Akurasi dan Konsistensi:** Setiap data, mulai dari Tipe Tiket hingga Prioritas dan Waktu Resolusi, harus akurat dan konsisten. Tipe dan ID harus saling mencocokkan, dan prioritas harus mencerminkan urgensi sebenarnya.
- **Kelengkapan Data:** Tidak boleh ada sel yang kosong. Semua informasi yang relevan harus dicatat untuk memungkinkan analisis yang komprehensif.
- **Standardisasi:** Ejaan dan penamaan harus distandardisasi. Contohnya, nama agen atau departemen harus memiliki format yang seragam untuk menghindari duplikasi data.
- **Realisme:** Nilai-nilai numerik seperti Waktu Resolusi harus realistis dan dapat digunakan untuk mengukur kinerja.

4. Create a one or two paragraphs to report to your manager.

JAWAB:

Berdasarkan tinjauan terhadap dataset tiket layanan TI, saya telah mengidentifikasi beberapa masalah kualitas data yang signifikan, termasuk inkonsistensi antara ID dan tipe tiket, prioritas yang tidak sesuai, dan sejumlah besar data yang hilang atau tidak logis, terutama pada Waktu Resolusi. Masalah-masalah ini dapat menghambat akurasi laporan kami dan menyebabkan kesimpulan yang salah mengenai kinerja tim *helpdesk*.

Untuk meningkatkan keandalan data, saya telah melakukan perbaikan manual pada setiap baris untuk mengoreksi tipe tiket, mengisi data yang hilang, dan menyesuaikan prioritas serta waktu resolusi agar lebih masuk akal dan konsisten. Sebagai langkah ke depan, saya merekomendasikan implementasi validasi data otomatis dan protokol pencatatan yang lebih ketat di sistem *helpdesk* untuk memastikan data yang dicatat akurat sejak awal.

CHALLENGE 3

Closed book, no GenAI is allowed.

Creating and Closing Incidents & Requests — Bank Nusantara Digital

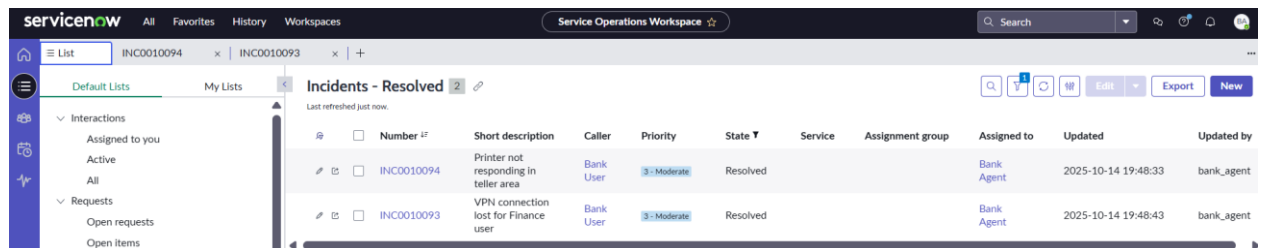
You are acting as a Service Desk Agent for Bank Nusantara Digital, an Indonesian banking company. You will create, process, and close two incidents and two service requests using your ServiceNow PDI, following ITIL best practices.

1. Log in to your ServiceNow PDI as admin.
2. Create two users:
 - Username: **bank_user** → role: user (End User / Customer). Set the password as: **Yeswecan!7**
 - Username: **bank_agent** → role: itil (Service Desk Agent). Set the password: **Yeswecan!7**
3. Create **two incidents** and **two service requests** for Bank Nusantara Digital with distinct categories:
 - Incident 1 → *Network* (e.g., ATM offline)
 - Incident 2 → *Hardware* (e.g., laptop won't power on)
 - Request 1 → *Software/Application* (e.g., SPSS installation)
 - Request 2 → *Account/Access* (e.g., VPN access request)
4. Work on each ticket as bank_agent and resolve/fulfill them.
5. Log back in as bank_user, review and close all tickets.
6. Submit:
 - Link to your PDI (make sure you use the predefined usernames and password)
 - Screenshots of:
 1. Two incidents from the agent's window (After resolution)
 2. Two requests from the agents' window (After resolution)
 3. A window of all closed tickets from user's window (After closure)

Link to your PDI: <https://dev305262.service-now.com/navpage.do>

Screenshots:

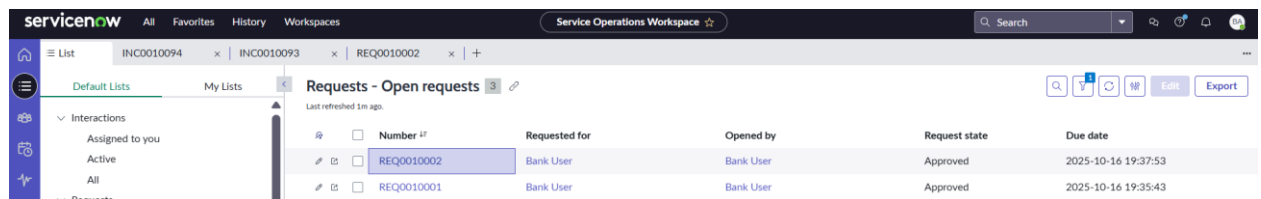
1. Two incidents from the agent's window (After resolution)



This screenshot shows the 'Incidents - Resolved' view in the ServiceNow interface. The left sidebar contains navigation options like 'Interactions', 'Assigned to you', 'Active', 'All', 'Requests', 'Open requests', and 'Open items'. The main table lists resolved incidents with columns for Number, Short description, Caller, Priority, State, Service, Assignment group, Assigned to, Updated, and Updated by.

Number	Short description	Caller	Priority	State	Service	Assignment group	Assigned to	Updated	Updated by
INC0010094	Printer not responding in teller area	Bank User	3 - Moderate	Resolved			Bank Agent	2025-10-14 19:48:33	bank_agent
INC0010093	VPN connection lost for Finance user	Bank User	3 - Moderate	Resolved			Bank Agent	2025-10-14 19:48:43	bank_agent

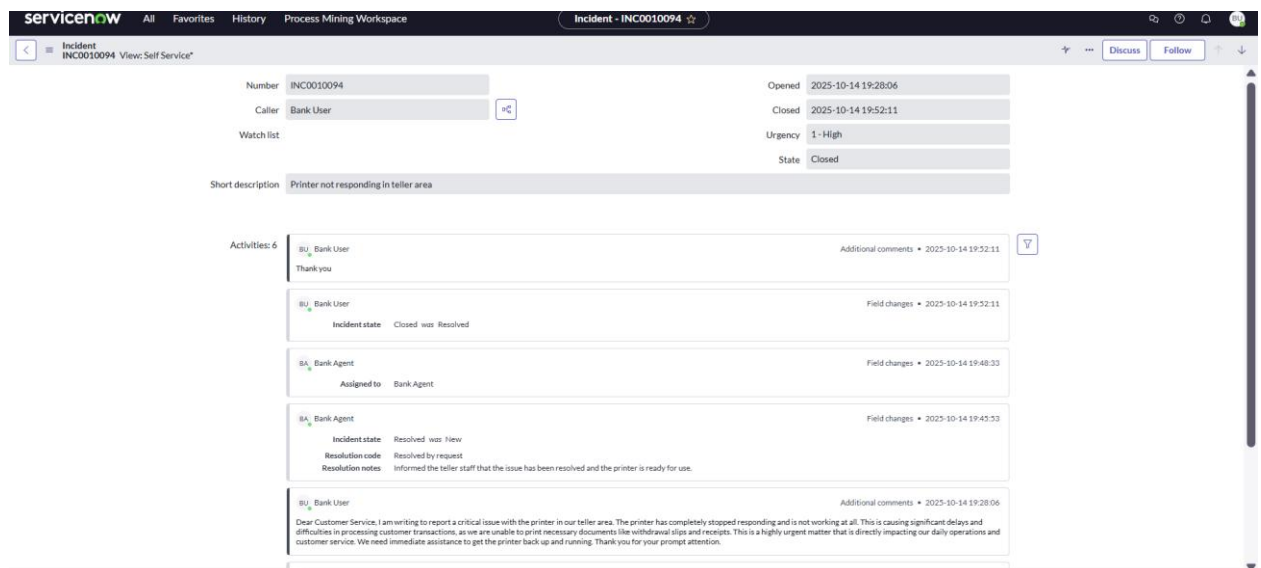
2. Two requests from the agents' window (After resolution)



This screenshot shows the 'Requests - Open requests' view in the ServiceNow interface. The left sidebar is the same as the previous screenshot. The main table lists open requests with columns for Number, Requested for, Opened by, Request state, and Due date.

Number	Requested for	Opened by	Request state	Due date
REQ0010002	Bank User	Bank User	Approved	2025-10-16 19:37:53
REQ0010001	Bank User	Bank User	Approved	2025-10-16 19:35:43

3. A window of all closed tickets from user's window (After closure)



This screenshot shows the detailed view of incident INC0010094. It includes fields for Number, Caller, Watch list, Short description, and a timeline of activities. The incident is in a 'Closed' state.

Number: INC0010094
Caller: Bank User
Watch list:
Short description: Printer not responding in teller area

Opened: 2025-10-14 19:28:06
Closed: 2025-10-14 19:52:11
Urgency: 1 - High
State: Closed

Activities: 6

- Bank User: Thank you (Additional comments: 2025-10-14 19:52:11)
- Bank User: Incident state Closed was Resolved (Field changes: 2025-10-14 19:52:11)
- Bank Agent: Assigned to Bank Agent (Field changes: 2025-10-14 19:48:33)
- Bank Agent: Incident state Resolved was View, Resolution code Resolved by request, Resolution notes Informed the teller staff that the issue has been resolved and the printer is ready for use. (Field changes: 2025-10-14 19:45:53)
- Bank User: Dear Customer Service, I am writing to report a critical issue with the printer in our teller area. The printer has completely stopped responding and is not working at all. This is causing significant delays and difficulties in processing customer transactions, as we are unable to print necessary documents like withdrawal slips and receipts. This is a highly urgent matter that is directly impacting our daily operations and customer service. We need immediate assistance to get the printer back up and running. Thank you for your prompt attention. (Additional comments: 2025-10-14 19:28:06)

servicenow

AllFavoritesHistoryProcess Mining Workspace

Incident - INC0010093

Incident

INC0010093 View: Self Service*

DiscussFollow

Number

INC0010093

Opened

2025-10-14 19:25:21

Caller

Bank User

Closed

2025-10-14 19:52:19

Watch list

Urgency

1 - High

State

Closed

Short description

VPN connection lost for Finance user

Activities: 6

Bank User

Additional comments • 2025-10-14 19:52:19

Thanks

Bank User

Field changes • 2025-10-14 19:52:19

Incident state Closed was Resolved

Bank Agent

Field changes • 2025-10-14 19:48:43

Assigned to Bank Agent

Bank Agent

Field changes • 2025-10-14 19:47:57

Incident state Resolved was New
Resolution code Resolved by request
Resolution notes Shared a knowledge article with the user on how to run a self-healing script for minor VPN client issues in the future.

Bank User

Additional comments • 2025-10-14 19:25:21

Dear Customer Service, I am writing to report a critical issue with my VPN connection. I've been unable to connect to the office network since this morning, which is preventing me from accessing important files and applications needed for my work in the Corporate Finance department. I have already tried restarting my laptop and reinstalling the VPN client, but the problem persists. This issue is highly urgent as it completely stops me from performing my duties. Could you please provide immediate assistance? Thank you for your help.